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Ferrous and Non-Ferrous metals

INTRODUCTION



- **Metals**: Exist in nature as **Ores** (compounds like oxides, carbonates, sulphides & phosphates)
- Metals used for Engineering purpose are classified as: **Ferrous and non-ferrous**

1. Ferrous Metals:

- Main constituent – **Iron** used as engineering material.
- **Magnetic** in nature & little resistance to corrosion

Stainless Steel



Alloy Steel



Wrought Iron



Cast Iron



Carbon Steel



Mild Steel



INTRODUCTION



2. Non Ferrous Metals:

- Do not contain iron (Non magnetic)
- High resistance to corrosion
- Metals other than iron like aluminium, copper, zinc, lead and tin.



IRON

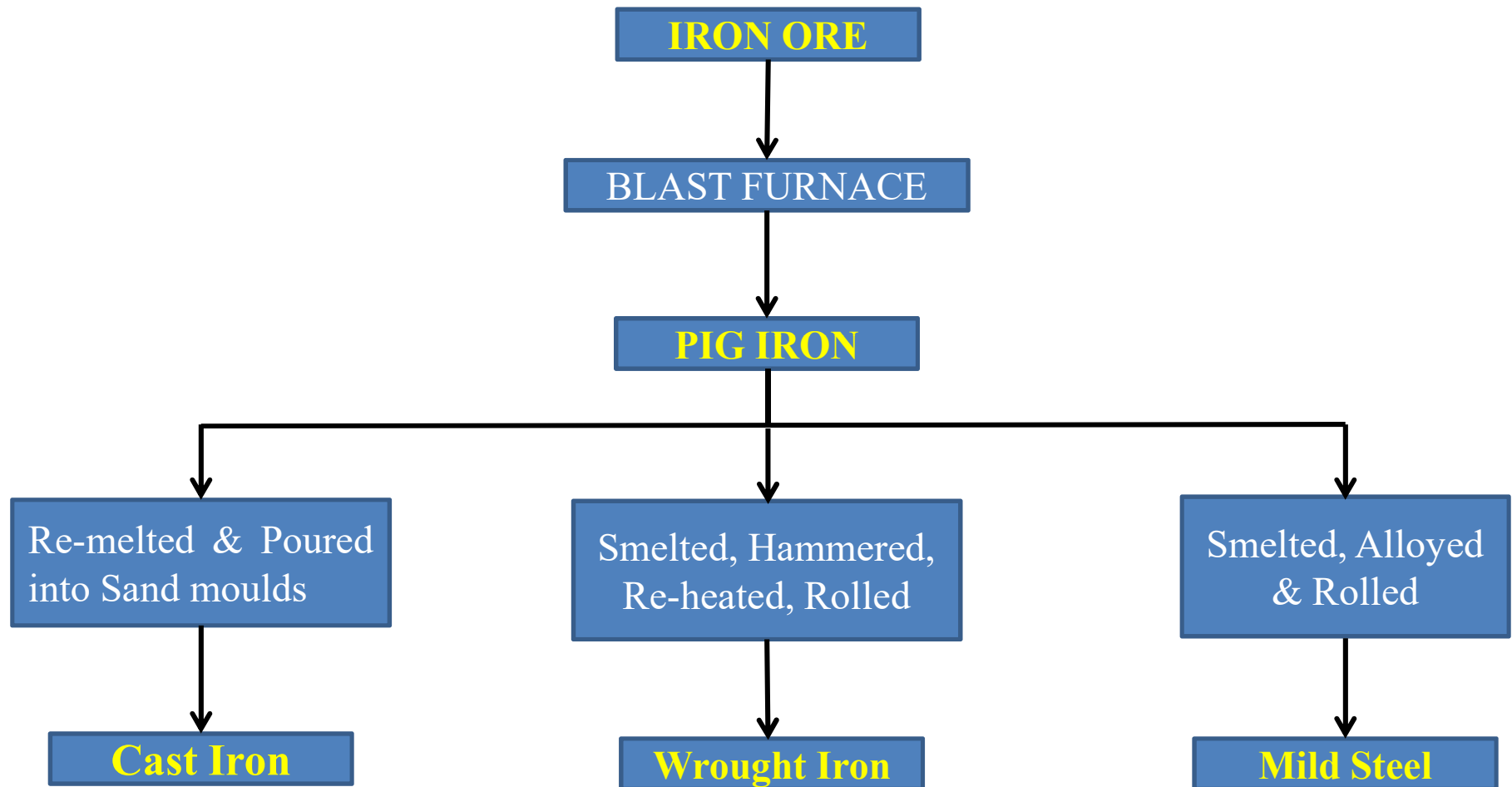


- Most used metal in Engineering Construction.
- Available in abundance but only in ore form (not free form)

Iron content of main ores are as follows:

- Magnetite (Fe_3O_4) — 70–75%,
 - Haematite (Fe_2O_3) — 70%,
 - Limonite ($2\text{Fe}_3\text{O}_3 \cdot 3\text{H}_2\text{O}$) — 60%, it is hydrated haematite,
 - Iron pyrite (FeS_3) — 47%, and
 - Siderite (FeCO_3) — 40%
-
- Iron has great utility because it alloys freely with other elements, and its inherent properties are markedly altered and improved for various applications

IRON contd..



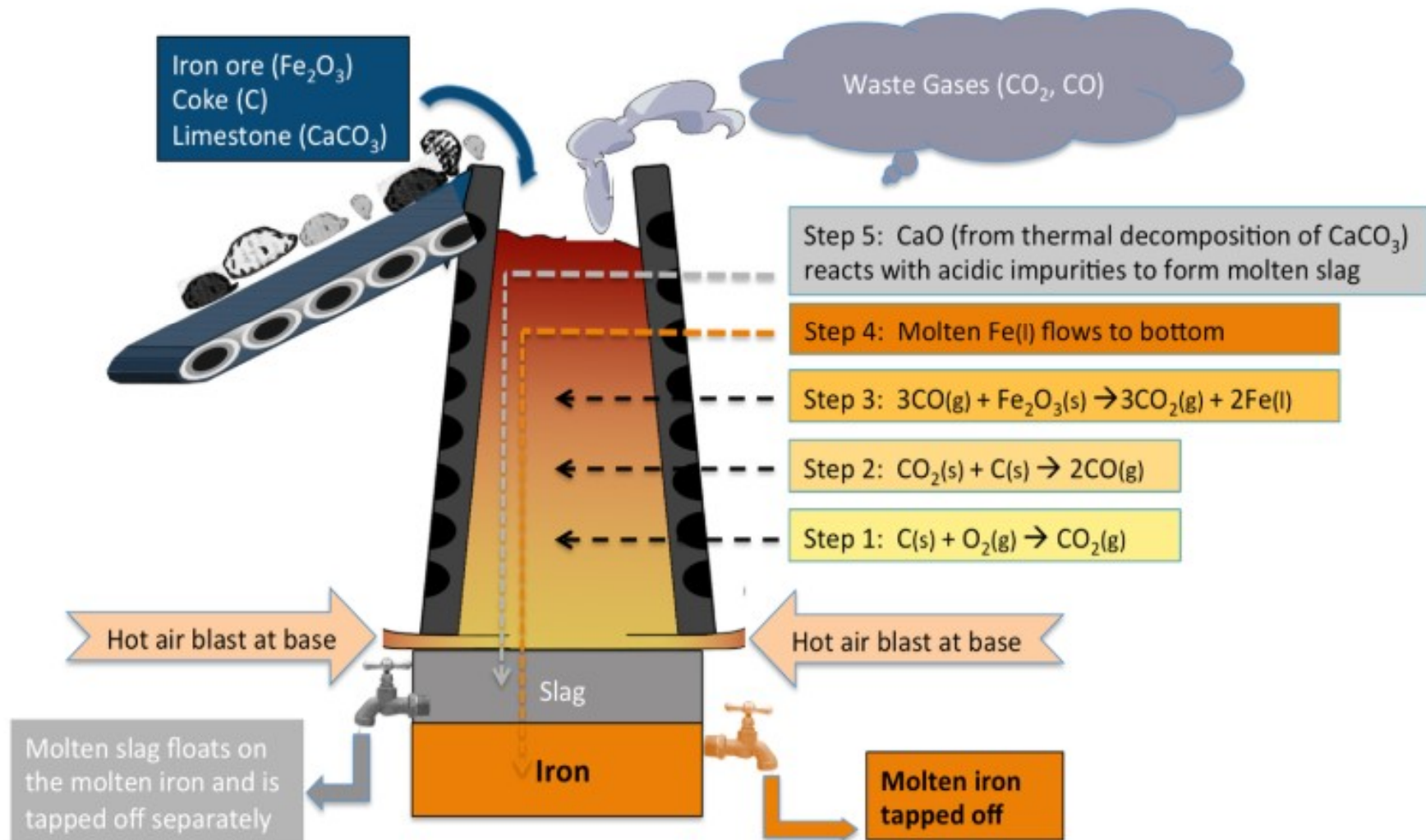
IRON



Pig Iron:

- Contains 3–4% carbon, 0.5–3.5% silicon, 0.5–2% manganese, 0.02–0.1% sulphur and 0.03–1% Phosphorus
- High carbon content i.e., **Hard & Brittle**
- **High** Compressive strength & **Weak** in Tension & Shear
- **Corrosion resistant**
- Cannot be welded or riveted
- **Uses:** **Columns, Base plate, door brackets** etc.,

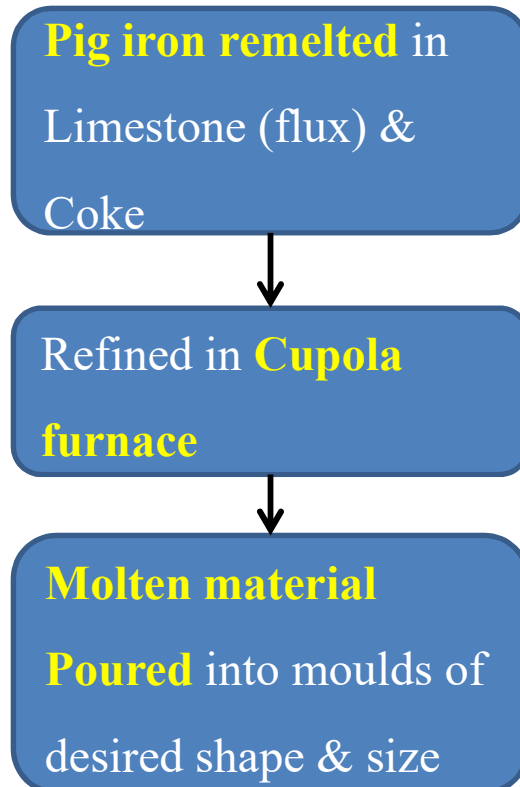
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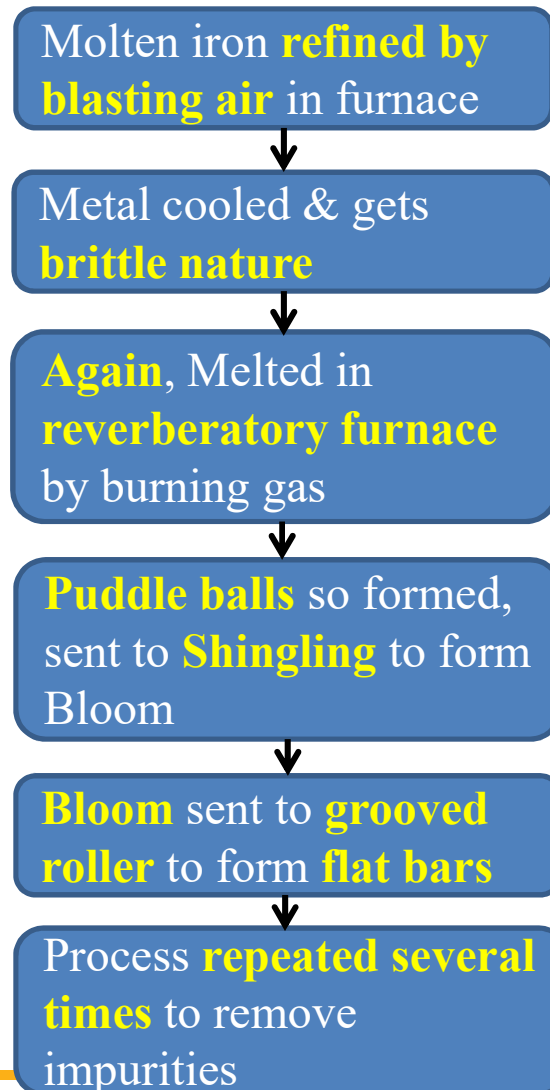


CAST IRON



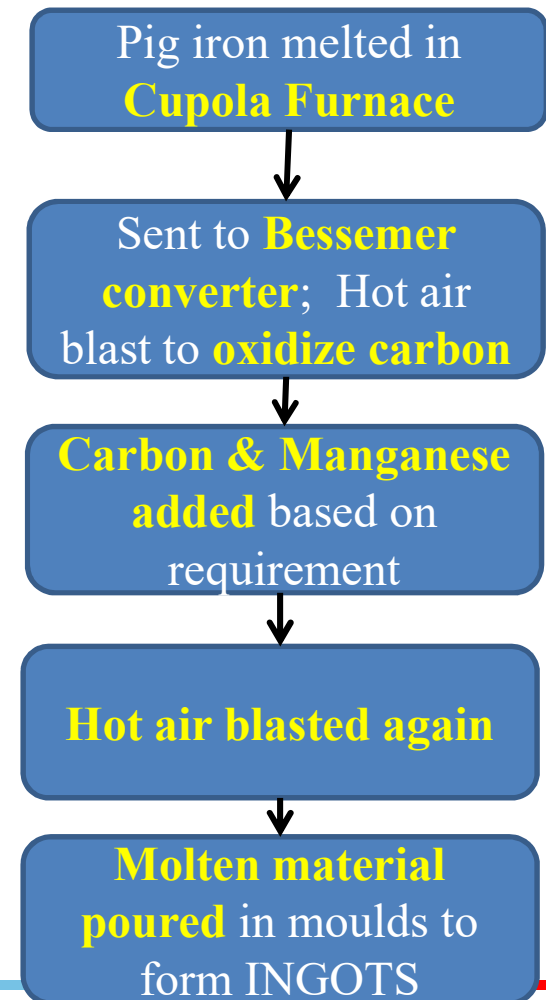
<https://www.youtube.com/watch?v=yVZJP3u9Kek>

WROUGHT IRON



MILD STEEL

Bessemer Process





IRON contd...

Properties	CAST IRON	WROUGHT IRON	MILD STEEL
Carbon Content	High	Low	Low
Nature	Hard & Brittle	Tough, malleable & Ductile	Tough, malleable & Ductile
Compressive Strength	600 MPa	200 MPa	250 MPa
Tensile Strength	150 MPa	240 MPa	840 MPa
Specific Gravity	7.50	7.80	8.00
Melting Point	1200°C	1500°C	1600°C
Weldability	Cannot be welded	Easily welded	Rapidly Welded
Reaction to Sudden Shock	Cannot withstand heavy shocks	Does not absorb shocks	Absorbs Shocks
Uses	Metal staircases, Sanitary fittings, gates etc.,	Nails, nuts, bolts, wires, roof coverings, rivets, chains, ornamental works etc.,	Structural steel, Machinery parts, Cookware, Pipelines etc.,

IRON ...contd



Cast Iron:

Methods of Casting

<u>Sand Casting</u> (Most common process)	* Wooden moulds in Packed Sand * Removed moulds after drying out Sand & cavity made. * Two holes made - One hole - Pouring metal & Another hole - Letting hot gases out * After Cooling ; Cast material taken out by breaking mould
<u>Hollow Casting</u> (Columns & Piles)	* Core placed to maintain hollow space * After cooling, core taken out and hollow cast iron obtained

IRON ...contd



Cast Iron:

Methods of Casting

<u>Centrifugal Casting</u> (Large Dia pipes, Gun barrels etc.)	* Molten material is poured in a revolving metallic cylindrical mould (10000 rpm) * Remove blow holes surface * Dense & Fine grained uniform structure
<u>Die Casting</u>	* Casting done under pressure(140 MN/mm²) * Removed early as water cooled

IRON ...contd



Cast Iron:

Effect of Impurities

Carbon	* As Carbon content↑ Melting temperature & Shrinkage↓
Silicon	* Fluidity↑ * Decreases blow holes formation↓
Sulphur	* Undesirable element as forms sulphides (with Fe & Mn) * Gets BRITTLE & WEAK at high temperatures * Neutralised by adding Silicon
Phosphorus	* Generally - 0.5% to 1.5% & >2% - Gets Brittle nature * Removes blow holes on surface * Dense & Fine grained uniform structure
Manganese	* General 0.4% - 1.2% * Tensile strength, hardness & shrinkage of iron↑



IRON ...contd

STEEL

- Most suitable building material among metallic materials
- Desired combination of hardness, ductility & strength can be obtained by controlling the carbon content, alloying elements and heat treatment.

Properties:

- Carbon ↑ ; Degree of hardness and strength ↑ ; Malleability & Ductility ↓ ;
Resistance to heavy shocks ↓
- Modulus of Elasticity – Independent of Carbon content

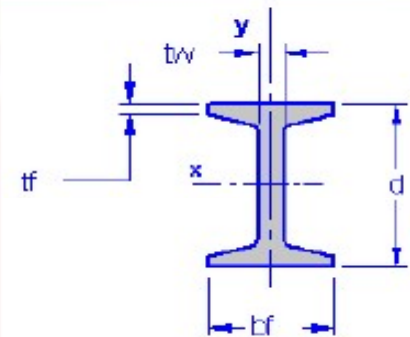
<https://www.youtube.com/watch?v=kSp5K70cYc>

IRON ...contd

STEEL

Major applications of steel

- Rolled sections
- Plates
- Fasteners
- Reinforcing bars
- Cables (Parallel wire & twisted wire)





IRON ...contd



STEEL

Manufacture Methods

- **Bessemer Process:** As explain earlier
- **Lintz and Donawitz (L.D.) process:** Modified process of Bessemer process ; **Temperature uncontrolled** ; Steel forms within **25 minutes**
- Other processes for steel manufacture:
 - Cementation process
 - Crucible process
 - Open Hearth process
 - Electric Smelting process
 - Duplex process

IRON contd...



STEEL

Properties	MILD STEEL	HIGH CARBON STEEL	HIGH TENSILE STEEL
Carbon Content	0.05 to 0.25% (Low)	0.55 to 1.50% (High)	0.60 to 0.80% (Medium)
Nature (Tough, malleable & Ductile)	Elastic than wrought iron	Elastic than Mild Steel	
Compressive Strength	800 – 1200 MPa	1350 MPa	
Tensile Strength	600 – 800 MPa	1400 – 2000 MPa	2000 MPa
Specific Gravity	7.30	7.90	8.00
Uses	Rolled sections, reinforcing bars , roof coverings and sheet piles and in railway tracks	RCC and Prestressed concrete members Can take Shocks ; Used in machine parts	Prestressed concrete construction

IRON ...contd



STEEL

Heat Treatment:

- **Objective** : To develop desirable properties in Steel
- It **affects** the nature, amount, and character of the metallographic properties like **Ductility, toughness, softness** etc.,
- Influences the solubility relationship b/w constituents
- The principal purpose of heat treatment are as follows:
 1. To **enhance properties** such as strength, ductility, hardness and toughness.
 2. To **relieve internal stresses and strains**.
 3. To **refine the grain**.
 4. To **remove gases**.
 5. To **normalize steel** after heat treatment.

Hardening, tempering, annealing, normalizing

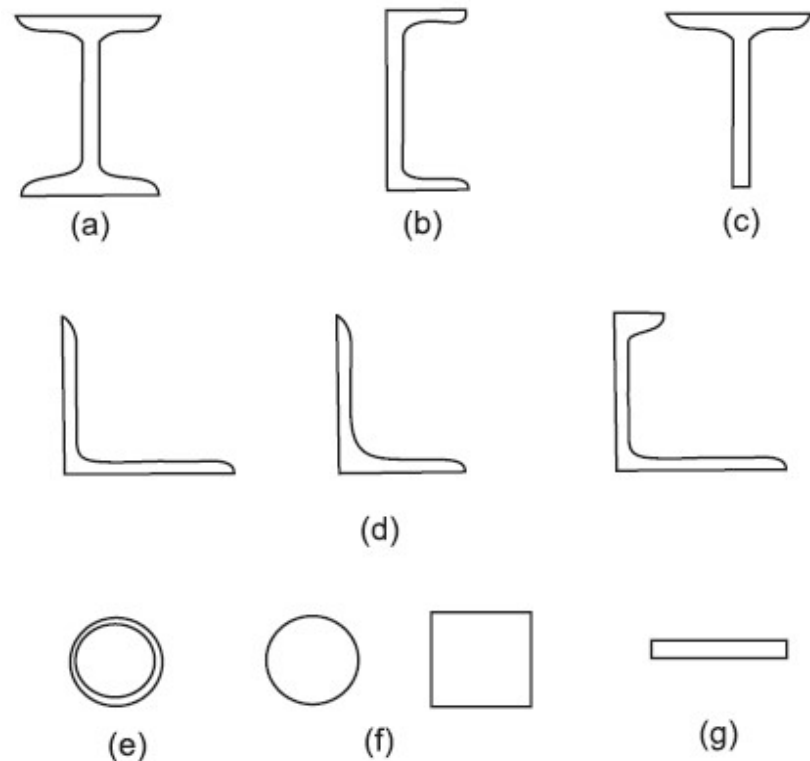
IRON ...contd



STEEL

Rolled Steel Sections: Structural steel can be rolled into various shapes and sizes

1. Rolled steel **I-sections** (a)
2. Rolled steel **channel sections** (b)
3. Rolled steel **T-sections** (c)
4. Rolled steel **angle-sections** (d)
5. Rolled steel **tube-sections** (e)
6. Rolled steel **bars** (f)
7. Rolled steel **flats** (g)
8. Rolled steel **plates**
9. Rolled steel **sheets**
10. Rolled steel **strip**



IRON ...contd



STEEL

Rolled Steel Sections Designation:

Rolled Steel Section Types	Criteria	Example
I-sections, Channel sections & T-sections	Depth and Self weight	I.S.L.B. 500 @ 735.7 N/m - "Indian Standard Light Weight Beam " Depth - 500mm & Self weight –735.7 N/m
Angle-sections	Leg lengths & Leg thickness	I.S.A. 40 × 25 × 6 mm - "Indian Standard Angle " - legs lengths - 40 mm and 25 mm & leg thickness - 6 mm
Flats & Strips	Width & Thickness	30 ISF 10 - "Indian Standard Flats " - 30 mm Wide and 10 mm Thick
Plates & Sheets	Length, Width & Thickness	I.S.P.L.2000 mm × 1000 mm × 8 mm - "Indian Standard Plate " - L x B x D
Tube-sections	Based on Outside Diameter & Self weight	

IRON ...contd



STEEL

Reinforcing Steel Bars:

- Concrete – **Weak in Tension** ; Requires Reinforcement
- **Good Steel** : Carbon content $< 0.25\%$, Sulphur $< 0.05\%$ & Phosphorus $> 0.05\%$
- Heat Treatment – Strength & mechanical properties $\uparrow$ & Steel requirement $\downarrow$

Type & grade of reinforcing bars : Based on Yield stress & Min. Elongation

- Mild Steel grade I (250 , 23)
- Mild Steel grade II (225, 23)
- Medium Tensile Steel (350, 20)
- High Yield Strength Deformed Steel bars (HYSD Bars) (415, 500; 14.5, 8)
- Thermo Mechanically Treated bars (TMT Bars) (415, 500; 22, 20)

IRON ...contd



STEEL

1. Mild Steel Bars :

- Designated as Fe- 250 (i.e., Yield Strength – 250 MPa)
- **High Ductility** - Best for impacts and suddenly applied loads

2. High Yield Strength Deformed Bars (HYSD Bars) :

- Do not have specific Yield point

After heat treatment, hot steel rods are either **heat rolled or cold twisted**

Has **deformations on surface**, ensures **better bond** between reinforcement & concrete.

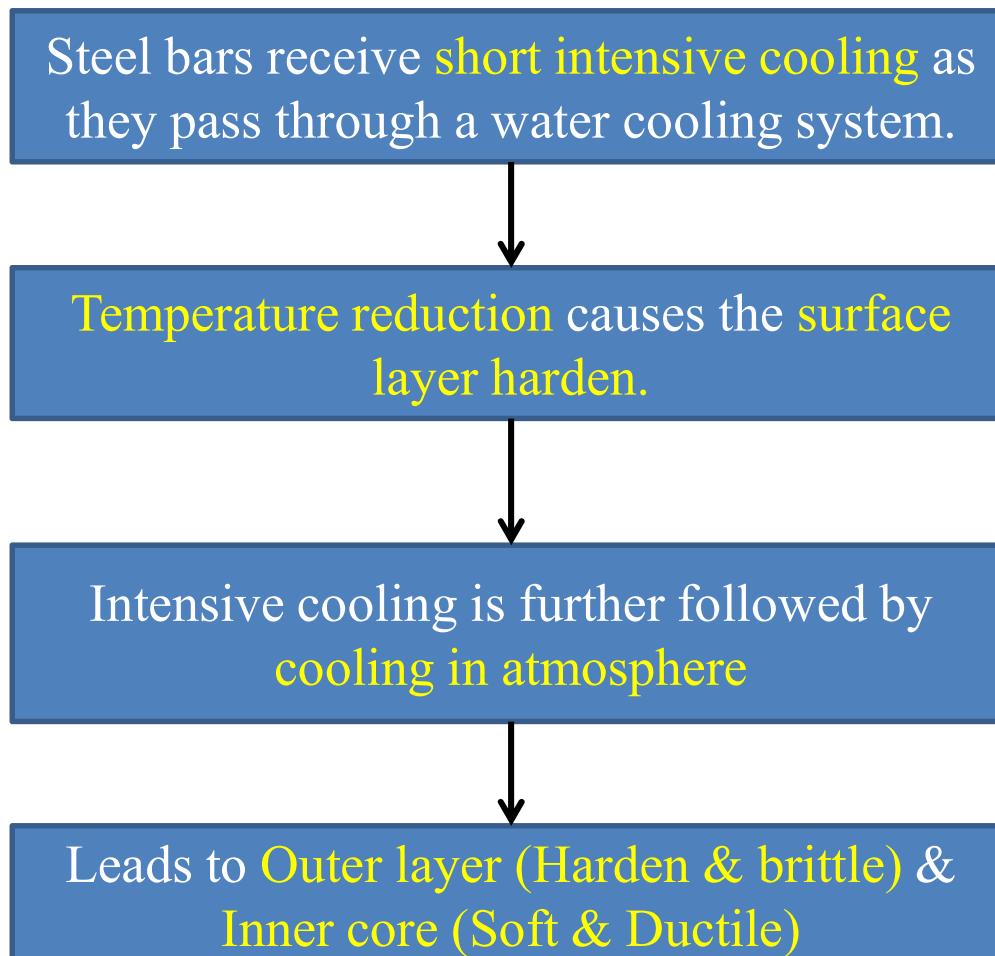
- Yield, tensile and bond strength ↑ when twisted hot or cold.
- **Cold twisted** deformed bars – **widely used in India** – Also called **TOR Steel**

IRON ...contd



STEEL

3. Thermo-Mechanically treated bars : Extra high strength reinforcing bars



IRON ...contd



STEEL

Properties of TMT Bars:

Resist high temperature (**up to 500°C**)

More ductile than CTD bars (HYSD) & good weldability

High fatigue resistance on dynamic loads

Highly recommended in **High Rise buildings**

Recent Advancement:

*Thermo-mechanically treated **high-strength corrosion resistant** (TMT-HCR) rebar*

- ✓ Unique **combination of strength, ductility and bendability**
- ✓ Superior resistance to aggressive weather conditions (**corrosion** effects)
- ✓ **Higher thermal resistance up to 600°C.**
- ✓ Used in **bridges, flyovers, dams, industrial structures, high rise buildings and underground platforms.**

IRON ...contd



STEEL

RUSTING & CORROSION:

Leading to oxidation when exposed to atmosphere (rain)

Methods to safeguard steel from above effect are:

1. **Enamelling** – Melting **flux** on surface of iron & coating with 2nd layer of more **fusible glaze**
2. **Galvanizing** – Coating iron with **thin film of zinc**
3. **Electroplating** – Metal coating such as **Chromium, nickel etc. on surface**
4. **Painting** – Applying a coat of **red lead** and then a coat of **enamel** or **aluminum paint**.

IRON ...contd



STEEL

TENSILE TEST ON STEEL:

To determine mechanical properties (E , f_y , σ_t , δl , μ etc.,)

PROCEDURE:

1. Specimen : Either rectangular ($W:T = 8:1$) or circular cross section
2. Gauge length (length $[L_0]$ on which specimen tested) is marked on the specimen

$$L_0 = 5.65\sqrt{A} \quad (A - \text{Cross-sectional Area})$$

3. Specimen gripped on either end in testing machine
4. It exerts axial pull so that specimen is stretched until it breaks

$$\text{Ultimate stress} = \text{Ultimate load} / \text{Area}$$

$$\text{Percentage Elongation} = \frac{(\text{Elongated length between gauge point} - L_0)}{L_0} \times 100$$

IRON ...contd



STEEL

TENSILE TEST ON STEEL:

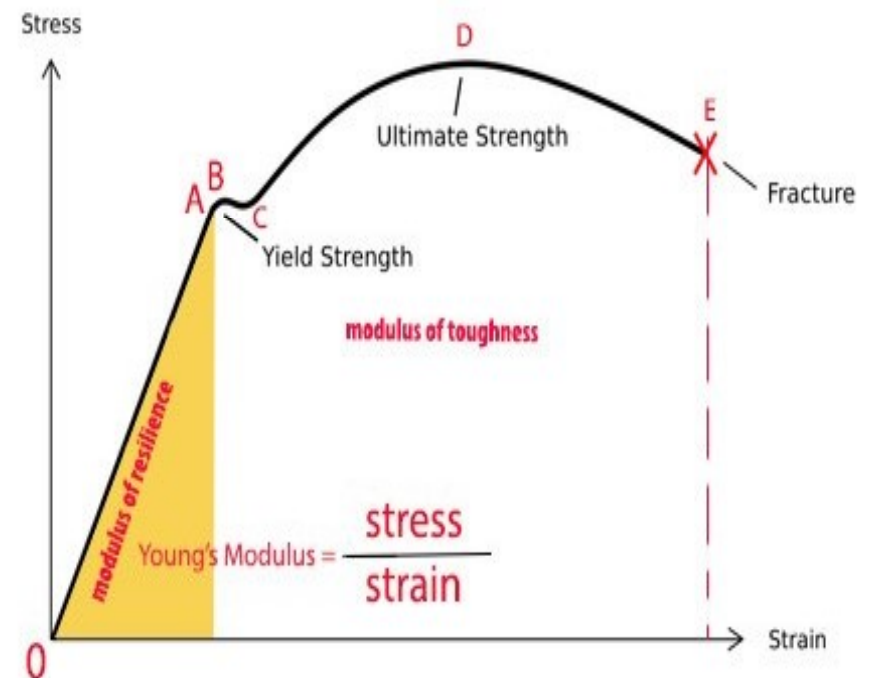
OA – Elastic Limit : Max. Stress at which **Stress $\propto$ Strain** (Hooke's law). Body **regain its original shape if load removed**.

B – Upper & C – Lower Yield Points

Yield Point: Stress at which strain increases without increase in load

D – Ultimate strength (Maximum value of stress in plastic region)

E – Breaking point : Point at which the body breaks



IRON ...contd

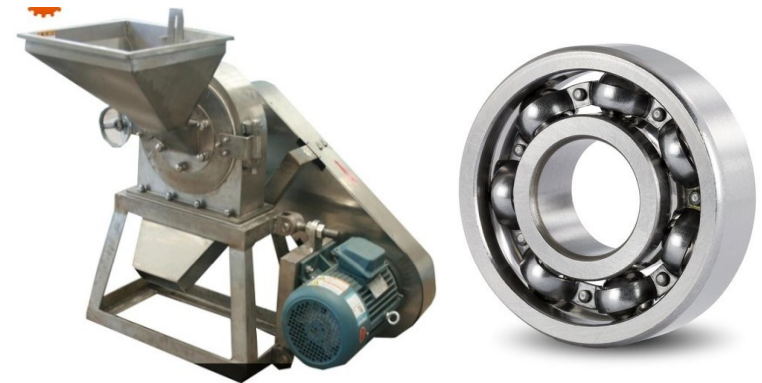


STEEL

ALLOY STEEL:

Combination of metals called **Alloys** are formed generally to attain desired properties

Alloy Steel	Composition
Stainless Steel	Chromium 16%
Nickel Steel	Nickel 3.5%
Invar Steel	Nickel 30 - 40%
Tungsten Steel	Tungsten 14 - 20%



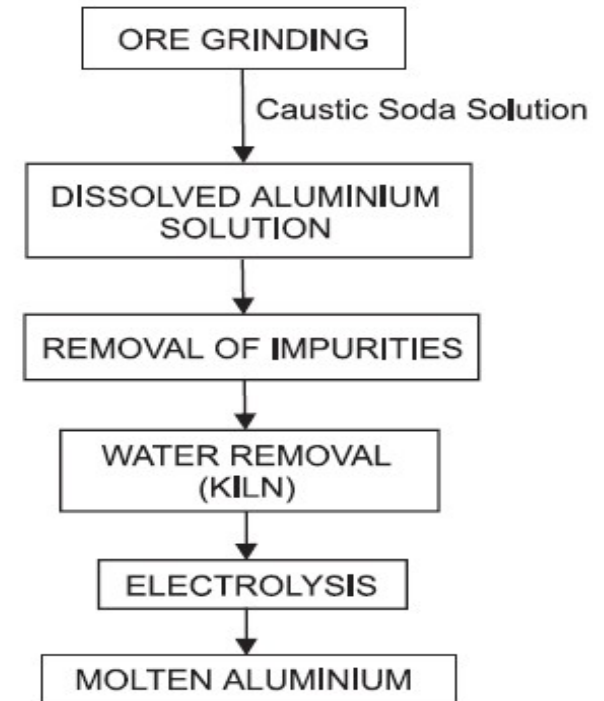
NON FERROUS MATERIALS



- Do not have any iron at all
- Despite high cost used in many engineering structures & industrial purposes

1. ALUMINIUM

- Primary ores: **Bauxite, corundum, kaolin**
- Ore purified by Bayer's process
- Reduced to Aluminum by Hall Hiroult's process



**Flow Diagram for
Extraction of Aluminium**

<https://www.youtube.com/watch?v=bl6OuGBx4nY>

Aluminium

Properties

Silver white

Very light but
strong & durable

Low thermal
conductivity, but
good conductor
of electricity

Malleable &
ductile

Can be riveted or
welded but not
soldered

Melting point
657°C, specific
gravity 2.7

Resistant to
nitric acid attack

Dissolves slowly
in Conc. H_2SO_4 &
soluble in HCl

Rapidly corroded
by caustic alkalis

Aluminium



Advantages

- Aesthetic appearance
- Corrosion resistant
- Very low maintenance cost
- Cost effective (Why?)
- Easy to fabricate, assemble, high scrap value

Pure Al is unsuitable for structural purposes

Alloyed

Uses

- Making door and window frames, Shop railings and corrugated sheets for roofing system
- Aluminum laminates, boards/panels to clad building facades
- Aluminium sheets - Over bathroom doors
- Aluminium powder – for making paint
- Automobile parts, airplane parts, packing for medicines, chocolates



Aluminium alloys

Some aluminium alloys are mentioned below:

<https://www.youtube.com/watch?v=t4HAzfhWtqE>

Alloy	Main alloying metal	Properties
Duralumin	Copper (3-5%)	Highly corrosion resistant Wire & sheets drawn; May be fabricated into different structural shapes for construction . Used in aircraft structures and other aerospace applications
Aldural	Coating of Al given to Duralumin	Better corrosion resisting properties

Magnalium, Y-alloy

NON FERROUS MATERIALS



2. COPPER

- Primary Ore: Copper pyrite, chalcopyrite etc., - **Smelting process.**

Properties:

Bright metal & Reddish Colour
Malleable & Ductile; Not weldable
Good conductor of heat & electricity " but " Resistivity increases with the impurities
High tensile Strength

Uses:

- Extensively used for **Electrical purpose and piping**
- Used as **roofing, sheeting & damp proofing** in Buildings

Brass: (Cu – 60-90% ; Zn - 10-40%)

Bronze: Alloy of **Copper & Tin**

NON FERROUS MATERIALS



3. ZINC

- **Main Source:** Sulphide ore zinc blende or black jack (ZnS , 67 % zinc).

Properties:

Corrosion Resistant
Good ductility & can be deformed in desired shapes
Durability and recyclability

Uses:

- Produce brass, German silver
- Corrosion resistive coating on iron (?), steel, boiler tubes, fruit jar covers and cans

NON FERROUS MATERIALS



4. LEAD

- Main Source: Lead Sulphide Ore, Galena (PbS , 86.6 per cent Lead).

Properties:

Colour: Blue grey &
Relative density- 11.34
Highly Malleable

Uses:

- In paints as base, **lead pipes and joints** in **sanitary fittings** and in **batteries**
- As **radiation shields**

NON FERROUS MATERIALS



5. TIN

- Main Source: Black oxide of tin, cassiterite (SnO_2 , 78.6 per cent tin).
- Extracted from ore by Crushing, Roasting & Melting up to 1000°C

Properties:

Colour: Silvery white , Lustrous & Highly Malleable , Soft (cut by knife)
Specific gravity – 7.3 ; Melting point- 232°C
Ductile as soft steel & stronger than lead.
High Corrosion resistance & low tensile strength

Uses:

- Tin coated sheets - used to make **cans, utensils & furnace pipes**
- Lead tin alloy – used for **roofing** ; making bronze & other alloys

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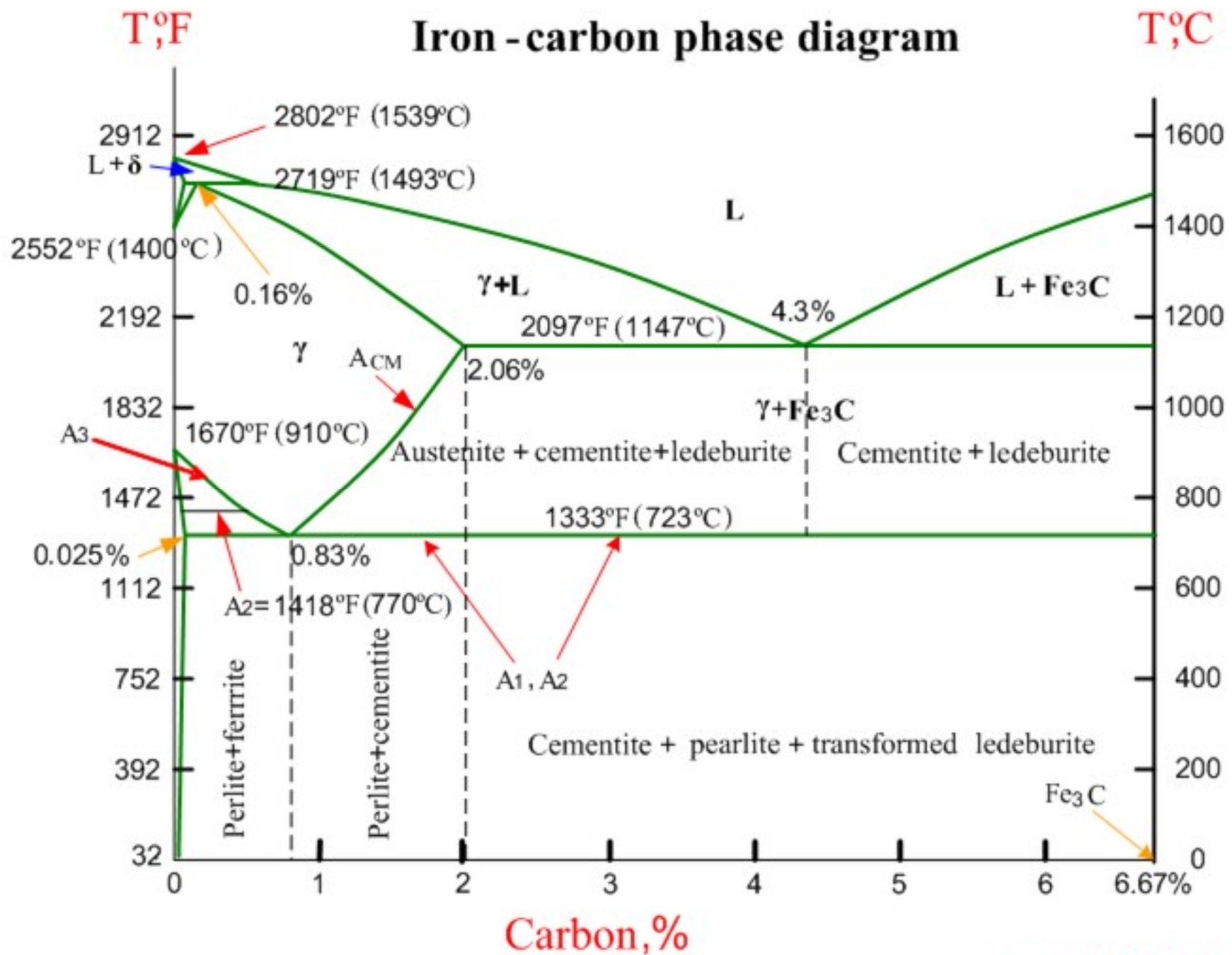
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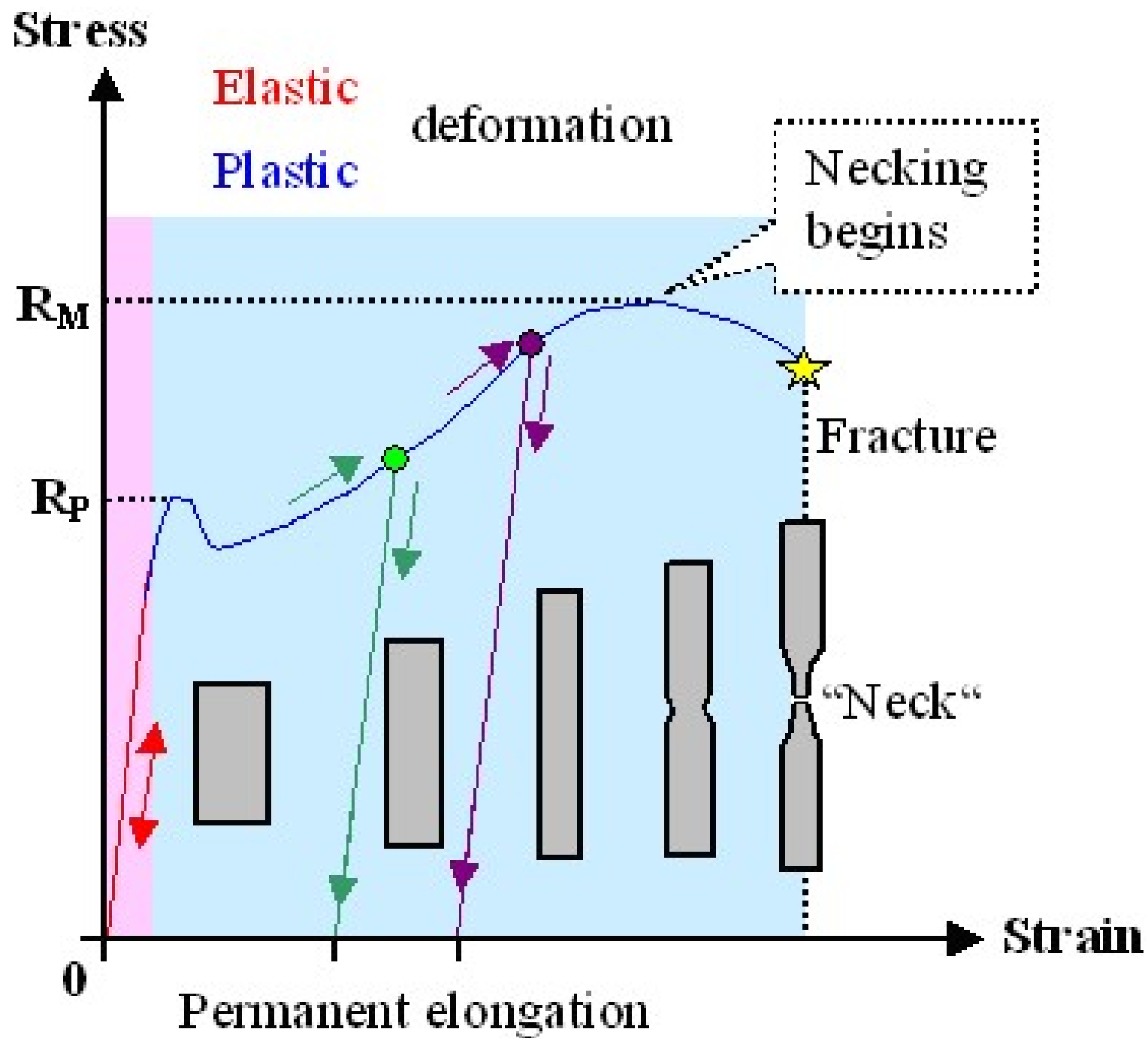


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THANK YOU

Iron - carbon phase diagram





Stress-Strain diagram for mild steel

